

3D Anatomy of Illness & Comprehensive Medical Dashboard

Interactive 3D Visual Diagnosis, Organ Fly-Through Simulation & Serial Lab Progression (April – October 2026)

● Critical Inpatient Status: Active Endocarditis & Sepsis

PATIENT NAME

Ram Kewal Mahato

UHID: CLH / 62 Y / Male

NEPHROLOGY DIAGNOSIS

CKD Stage G5

Thrice-Weekly Hemodialysis

CARDIOLOGY DIAGNOSIS

Infective Endocarditis

1.9 × 1.8 cm Mitral Mass (LVEF 40%)

NEUROLOGY DIAGNOSIS

Cardioembolic Strokes

Bilateral Cerebral + Cerebellar CVA

ORTHOPEDIC SOURCE

Left Hip Prosthesis PJI

Klebsiella species (SilverStreak)

SEPSIS SEVERITY

PCT 11.16 ng/mL

Severe Septic Shock (Normal <0.5)

INTERACTIVE 3D WEBGL VISUALIZATION



Click and drag anywhere on the 3D canvas to rotate around Papa's body in 360°. Scroll or pinch to zoom into any organ. Watch glowing bacteria rise from the left hip and clot fragments shoot up into the brain.





Stage 1: The Weakened Soil (Diabetic CKD G5)

15 Years of High Blood Sugar Damaged Both Kidneys

The illness begins with long-standing Type 2 Diabetes Mellitus and High Blood Pressure over 15 years. Over time, high glucose scarred the microscopic kidney filters. A kidney biopsy conducted at **GB Pant Hospital** proved that 50% to 60% of Papa's kidney tissue was permanently fibrosed (Kimmelstiel-Wilson nodular glomerulosclerosis). This created **End-Stage Renal Disease (CKD Stage G5)**, requiring permanent hemodialysis thrice weekly. This also caused persistent anemia (low hemoglobin) and severely suppressed the immune system's white blood cells, leaving the body with no defense against bacterial invasion.

SERUM CREATININE

6.35 mg/dL ↑

BLOOD UREA (PEAK)

205.4 mg/dL ↑

KIDNEY BIOPSY (GB PANT)

50-60% Fibrosis

DIALYSIS FREQUENCY

3 Times / Week

Why this mattered to the whole body: With the kidneys failed, uremic toxins paralyzed the white blood cells. When bacteria entered at the hip, the immune system was completely unable to contain them.

VISUAL PATHOPHYSIOLOGY



This medical crisis was not five unrelated sicknesses. It was a single, interconnected **biological domino effect** that began in the kidneys, ignited at the left hip surgical wound, invaded the bloodstream, seeded the heart valve, and finally showered blood clots into the brain.

STEP 1

FOUNDATION



Diabetic Kidney Failure (CKD G5)

15 years of high sugar and blood pressure permanently scarred 50–60% of both kidneys, creating zero physiological reserve and suppressing immunity.

GB Pant Biopsy: Kimmelstiel–Wilson Sclerosis | Urea 205 mg/dL

STEP 2

THE SPARK



Infected Left Hip Prosthesis

Left hip artificial joint developed deep infection. Resistant *Klebsiella* bacteria adhered to the metal, forming a protective biofilm shedding into veins.

SilverStreak Pus Culture: *Klebsiella* species Isolated

STEP 3

THE INVASION



Bloodstream Sepsis Shock

Bacteria flooded the bloodstream. The immune system panicked into septic shock, causing extreme inflammation and consuming blood platelets into clots.

CLH ICU Lab: PCT 11.16 ng/mL (Severe Shock) | Platelets 64K

STEP 4

THE TICKING CLOT



Mitral Valve Endocarditis

Bacteria attached to the heart's anterior mitral valve leaflet, multiplying into a massive 1.9 × 1.8 cm mobile vegetation and dropping pump strength to 40%.

Chandra Laxmi 2D Echo: 1.9 × 1.8 cm Vegetation | LVEF 40%

STEP 5

THE IMPACT



Embolic Strokes in the Brain

Because the mitral valve pumps directly to the head, pieces of the infected clot broke loose and blocked brain arteries, causing sudden confusion and strokes.

Brain MRI (30-Sep): Multi-Infarct CVA + Petechial Microbleeds

ANATOMICAL CLOSE-UPS

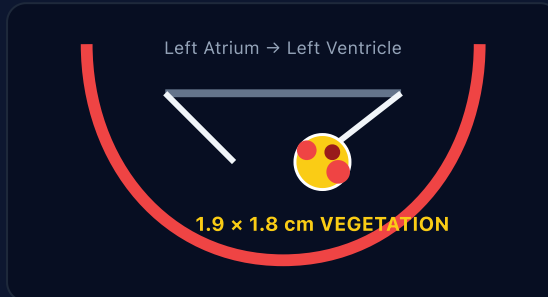


Detailed illustrated cutaways showing how the bacteria colonized the prosthetic hip, seeded the heart valve, launched strokes to the brain, and affected the scarred kidneys.



Mitral Valve Vegetation

Infective Endocarditis



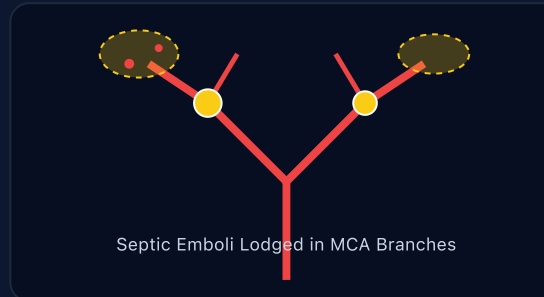
Bacteria from the blood adhered to the anterior leaflet of the mitral valve, forming a large **1.9 × 1.8 cm mobile vegetation** (an infected bacterial clot). This mass damaged valve closure (mitral regurgitation) and dropped the heart's pumping power (LVEF dropped to 40%).

Mass: 1.9 × 1.8 cm | LVEF 40% (CLH Echo)



Cardioembolic Strokes

Arterial Occlusion & Microbleeds



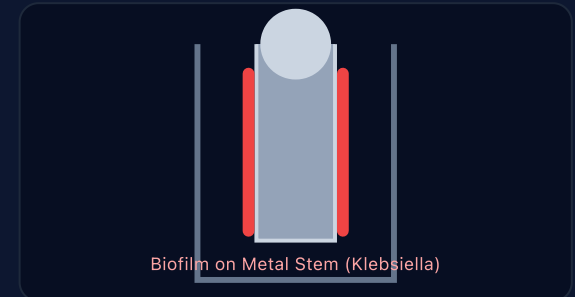
Because the heart valve pumps directly into the carotid arteries, pieces of the infected 1.9 cm vegetation broke loose and traveled up to the brain. These **septic emboli occluded cerebral vessels**, causing ischemic strokes and petechial microbleeds confirmed on Brain MRI.

MRI (30-Sep): Multi-Infarct CVA + SAH



Infected Hip Prosthesis

The Primary Bacterial Reservoir



Bacteria (**Klebsiella species**) colonized the metal stem of the left hip hemiarthroplasty. On metal implants, bacteria secrete a thick "biofilm" shield that oral antibiotics cannot penetrate, continuously shedding bacteria into the bloodstream.

SilverStreak Pus: Klebsiella Species



Scarred Kidneys (CKD G5)

Kimmelstiel-Wilson Sclerosis



50-60% Fibrosis & Scarred Glomeruli

GB Pant Hospital biopsy revealed classic **Kimmelstiel-Wilson nodular sclerosis** with 50% to 60% interstitial fibrosis and tubular atrophy. With natural filtration near zero, toxins and fluids build up unless removed by hemodialysis.

GB Pant Biopsy: 50-60% Fibrosis (CKD G5)



Clear, direct medical explanations for the questions every family member asks about this illness.

? How did an infection in his left hip reach his brain?

The circulatory system is a continuous highway. Bacteria from the left hip wound entered the femoral veins, traveled up the vena cava into the heart, and colonized the mitral valve. The heart valve then acted like a launcher: pieces of the infected mass (vegetation) broke loose into the arterial blood and were pumped straight up the carotid arteries into the brain, causing strokes.

? Why did his blood platelets drop to 64,000?

When severe bacterial sepsis occurs (Procalcitonin 11.16 ng/mL), the body's clotting system goes into overdrive. Platelets are consumed in thousands of microscopic clots throughout the blood vessels and inside the 1.9 cm heart vegetation itself. This rapid consumption causes the circulating platelet count to drop from normal (>150,000) down to 64,000 / μ L.

? Why can't doctors give strong blood thinners to clear the strokes?

This is one of the most dangerous medical dilemmas. The brain MRI showed **petechial microbleeds and trace subarachnoid hemorrhage** alongside the ischemic strokes. Giving strong blood thinners (like heparin or tPA) could cause those tiny bleeding spots to burst into massive, life-threatening brain hemorrhages. Therefore, therapy must focus on fighting the bacteria and stabilizing blood pressure.

? Why does Papa still need regular hemodialysis right now?

Because his kidneys have 50-60% permanent scarring from diabetes, they cannot excrete urea, creatinine, potassium, or fluid. On admission, his Blood Urea soared to **205.4 mg/dL** (causing uremic brain toxicity). Hemodialysis successfully cleared his urea down to **64.8 mg/dL**. Dialysis is life-preserving, but must be performed gently to prevent drops in blood pressure that could worsen the brain strokes.

QUANTITATIVE SEVERITY



Visualizing current ICU lab markers against safe physiological reference limits.

SEPSIS SEVERITY (PCT)

11.16 ng/mL



Safe: < 0.50 🚨 **Severe Septic Shock**

HEART EF% (LVEF)

40 % (Dropped from

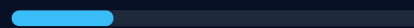
55%)



Normal: > 55% **Impaired Systolic Pump**

BLOOD PLATELET COUNT

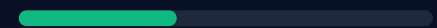
64,000 / μ L



Normal: 150K - 410K **Thrombocytopenia Alert**

AZOTEMIA (BLOOD UREA)

64.8 mg/dL (Post-HD)



Peak: 205.4 mg/dL **Cleared by Dialysis**

HEMOGLOBIN (HB)

8.5 g/dL



Normal: 13.0 - 17.0 **Chronic CKD Anemia**

MASTER CLINICAL RECORDS



Every test result across 13 serial testing dates from Chandra Laxmi Hospital, Dr Lal PathLabs, GB Pant Hospital, Janakpuri Super Speciality Hospital, Sir Ganga Ram Hospital, SilverStreak Hospital, and Thyrovision Labs.

 Complete Blood Count (CBC / Hemogram) Serial Tracking

Parameter	Reference Range	15-Apr (JSSH)	20-Apr (JSSH)	02-May (JSSH)	14-May (JSSH)	23-May (Silver)	03-Jun (Thyro)	08-Aug (Self)
Hemoglobin (Hb) [g/dL]	13.0 – 17.0	8.40 ↓	8.10 ↓	8.20 ↓	8.00 ↓	8.10 ↓	7.90 ↓	6.40 ↓↓
WBC Count (TLC) [/μL]	4,000 – 10,000	9,200	8,900	10,100 ↑	9,800	—	—	—
Platelet Count [/μL]	150,000 – 410,000	165,000	158,000	170,000	162,000	—	152,000	—
Neutrophils [%]	40.0 – 80.0	—	—	—	—	—	—	—
PCV / Hematocrit [%]	40.0 – 50.0	25.2 ↓	24.8 ↓	25.0 ↓	24.5 ↓	24.4 ↓	24.1 ↓	22.5 ↓

 Kidney Function Tests (KFT), Azotemia & Electrolytes

Parameter	Ref Range	15-Apr (JSSH)	02-May (JSSH)	28-May (Nephro)	03-Jun (Thyro)	05-Jun (SGRH)	28-Sep (Adm)	30-Sep (Post-HD)
Serum Creatinine [mg/dL]	0.60 – 1.20	5.80 ↑	4.90 ↑	4.30 ↑	4.40 ↑	4.30 ↑	6.35 ↑	3.78 ↑
Blood Urea [mg/dL]	17.0 – 49.0	142.0 ↑	118.0 ↑	103.0 ↑	106.0 ↑	103.0 ↑	205.42 ↑	100.62 ↑
Serum Sodium (Na+) [mEq/L]	136.0 – 145.0	138.0	137.0	136.0	—	—	137.5	136.8
Serum Potassium (K+) [mEq/L]	3.5 – 5.1	4.20	4.00	3.80	—	4.10	4.30	3.90

 Sepsis, Inflammation, Iron & Critical Diagnostics

Date	Facility	Test Name	Result	Reference Range	Clinical Significance
30-Sep-2026	Chandra Laxmi ICU	Serum Procalcitonin	11.16 ng/mL 🚨	< 0.50 ng/mL	Severe Septic Shock (>10 indicates massive bloodstream bacterial invasion)
28-Sep-2026	CLH Admission / Dr Lal	C-Reactive Protein (CRP)	113.84 mg/L ↑↑	< 6.00 mg/L	Severe systemic inflammatory surge on ICU presentation
02-Oct-2026 AM	Chandra Laxmi ICU	C-Reactive Protein (CRP)	74.88 mg/L ↑	< 6.00 mg/L	Partial reduction following IV antibiotic therapy

Date	Facility	Test Name	Result	Reference Range	Clinical Significance
02-Oct-2026 PM	Dr Lal PathLabs	hs-CRP (High Sensitivity)	93.30 mg/L ↑	< 3.00 mg/L	Persistent high-grade vascular & endocardial inflammation
03-Jun-2026	Thyrovision Labs	Serum Iron	24.90 µg/dL ↓↓	59.0 – 158.0 µg/dL	Profound absolute iron deficiency anemia in CKD G5
03-Jun-2026	Thyrovision Labs	Transferrin Saturation (TSAT)	9.02 % ↓↓	20.0 – 50.0 %	Indication for IV Ferinject 500mg infusion at Ganga Ram Hospital
22-Apr-2026	Janakpuri Super Speciality	Vitamin D (25-OH)	< 8.00 ng/mL ↓↓	30.0 – 100.0 ng/mL	Severe deficiency causing secondary hyperparathyroidism & bone weakness
25-Apr-2026	Janakpuri Super Speciality	Arterial Blood Gas (ABG pH)	7.31 ↓	7.35 – 7.45	Compensated metabolic acidosis of kidney failure (Bicarbonate 16.8)

THERAPEUTIC PLAN



Current multi-disciplinary treatment regimen managing sepsis, kidney failure, cardiac support, and stroke prevention.

Medication / Therapy	Dose & Route	Drug Class	Primary Indication	Response & Status
Inj. Monocef (Ceftriaxone)	1g to 2g IV Daily	3rd Gen Cephalosporin	Empirical broad-spectrum sepsis coverage	WBC decreased from 18,990 to 14,080 / μ L; culture tailoring ongoing
Inj. Forcan (Fluconazole)	100mg IV Daily	Antifungal Azole	Fungal prophylaxis in diabetic immunocompromised state	Prophylactic fungal protection maintained
Inj. Emset (Ondansetron)	4mg IV TDS / SOS	Antiemetic	Nausea from uremia and dialysis shifts	Symptomatic relief achieved
ICU Hemodialysis	Thrice Weekly (3.5 – 4.0 hrs)	Renal Replacement Therapy	End-Stage Renal Disease (CKD G5)	Urea successfully cleared: 205.4 $\rightarrow$ 64.8 mg/dL; Creatinine 6.35 $\rightarrow$ 3.06 mg/dL
Sliding Scale Regular Insulin	Subcutaneous as per GRBS chart	Antidiabetic Hormone	Type 2 Diabetes Mellitus	Tight blood glucose control avoiding hypoglycemia in kidney failure
Tab. Thyronorm (Levothyroxine)	25 mcg OD (Morning empty stomach)	Thyroid Hormone	Hypothyroidism	Daily baseline hormone maintenance
Tab. Nefrosave	1 tablet TDS post-meals	Alpha-Ketoanalogues	Uremic toxin control & nitrogen balance (Dr. D. S. Rana)	Ongoing chronic nephrology therapy
Tab. Shelcal-HD	1 tablet OD post-meals	Calcium Carbonate + Vit D3	Phosphate binding & bone mineral management in CKD	Baseline calcium maintenance

Confidential Family Medical Archive — Ram Kewal Mahato (62Y / Male) | Prepared with 3D anatomical WebGL visualization, clinical infographics, and verified records from Chandra Laxmi Hospital, Dr Lal PathLabs, GB Pant Hospital, Janakpuri Super Speciality Hospital, Sir Ganga Ram Hospital, SilverStreak Hospital, Artemis Hospital, Prognosis Laboratories, and Thyrovision Labs. Always consult the primary treating physicians for clinical decisions. Generated: October 05, 2026.